

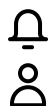
[skip to content](#)

This is a page that we are continually improving.

We would love to hear your feedback and suggestions.



International Journal of Biometeorology



[Account](#)

Are Heat-Health Systems Socially Blind? Social Isolation and Dehydration-Related Healthcare Utilization Across Japan

Action needed

You need to make revisions to your submission

Your reviewers have provided feedback on your submission. There are revisions you need to make before your submission can progress. We've also emailed m211039@fmu.ac.jp with more details.

We recommend submitting your revisions by 14 Aug 2026.

Feedback from the Editor

▼ Read editor's comments

Dear Dr Saito,

Your manuscript, "Are Heat-Health Systems Socially Blind? Social Isolation and Dehydration-Related Healthcare Utilization Across Japan", has now been assessed.

We invite you to revise your paper, carefully addressing the comments from the reviewers and the editor. Please ensure the results are accurately reported, any overstated conclusions are rewritten and the limitations of the work fully explained. When your revision is ready, please submit the updated manuscript and a point-by-point response. This will help us move to a swift decision.

We recommend submitting all revisions within the mentioned deadline.

If you need more time, please contact us and include your submission ID.

Kind regards,

Jennifer Fitchett

Field Editor

International Journal of Biometeorology

[Support contact: francisanthony.vega@springernature.com](mailto:francisanthony.vega@springernature.com)

Submission ID: 0cd18650-d5db-481f-b00b-bbde3cda74f7

Feedback from the Reviewer

▼ Read reviewer's comment

Reviewer 1

Thank you for the opportunity to review your manuscript. I think it was an interesting topic. The authors applied several interesting 'checks' to their analyses. I have commented in the document itself. My main concerns are that the relationship between heat exposure and dehydration may not be linear (not previous literature was mentioned about this exposure response function); AC in homes was high but it is unknown if it was used; and it was unknown what other health conditions older adults had that may also be associated with dehydration (older adults are also known for drinking less water). Still, I liked the discussion and I think with more literature in the introduction, additional details in the methods and a toned down 'association' between heat and therapy, this does present and interesting ecological study.

Attachment(s):

- [Download Reviewer 1 attachment 1](#)

Reviewer 2

This is an important and timely study using nationwide data from all 47 Japanese prefectures. The open data and code, sensitivity analyses, and clear reporting are strengths. However, the central interpretation needs further support, and I recommend

major revision.

1. Population age structure is the main concern. Infusion therapy for dehydration is likely to be concentrated among older people, but the outcome is calculated per 100,000 total population. Elderly solo household rate will also be related to the proportion of residents aged 65 years or older. The association may therefore show that older prefectures have more dehydration related treatment rather than an independent effect of social isolation. Please age standardise the outcome or calculate it using an elderly denominator. At minimum, include prefectural ageing rate alongside elderly solo household rate and show how the estimate changes. Collinearity does not remove the need to address this confounding.

2. Healthcare supply and regional treatment practices may also explain part of the association. Infusion use can reflect local service availability and clinical practice as well as morbidity. Please add a measure such as hospital beds or clinics per capita. If this is not feasible, discuss supply induced demand as a leading alternative explanation.

3. The stratified analysis needs fuller interpretation. The association is absent in the higher isolation stratum and concentrated in the lower isolation stratum. This does not support a simple dose response relationship. Please test for nonlinearity, examine the influence of metropolitan prefectures, and discuss whether the result may reflect confounding.

4. The claim that social factors outperform climatic indicators is not convincing with the current climate measures. Seasonal averages based on daily mean temperature may obscure the peak exposures most relevant to heat illness. Please use stronger measures, such as maximum based WBGT, days above a WBGT threshold, or heat degree days. If these cannot be added, remove the comparative claim from the title, abstract, and conclusions.

5. The claimed sixfold difference from the negative control is not valid because the

raw coefficients concern outcomes measured on different scales. The correlations and R^2 values show a much smaller difference. General outpatient use is also affected by ageing, access, and healthcare supply, so it may not be a suitable negative control. Please remove the sixfold claim, report standardised comparisons, and reconsider the interpretation of this analysis.

6. Infusion use is only a partial proxy for heat related illness. Its correlation with heatstroke ambulance transport is helpful, but infusions may be given for many other reasons and the data do not establish a summer peak. Please describe the outcome more cautiously as general dehydration related utilisation with a possible heat component, and temper the policy conclusions accordingly.

The study should also be framed as hypothesis generating given the number of univariate analyses and lack of adjustment for multiple comparisons. A descriptive title would better match the ecological design because no warning system is directly evaluated. Please also explain the mismatch among the 2014, 2020, and 2023 variables and correct the GitHub link containing “XXX”.

The paper could make a useful contribution if the association remains after adequate control for age structure and healthcare supply.

How to submit your revision

The original submitting author must upload a point-by-point response to the comments as a PDF file. This must include a description of any additional experiments that were carried out and a detailed rebuttal of any criticisms or requested revisions that you disagreed with.

Any files (including the manuscript) that have changed based on the comments will also need to be uploaded again. Do not include tracked changes in your manuscript file. If you need to upload a marked-up version of the revised manuscript with the changes highlighted you can upload it on the related file section.

Please note the original submitting author may be different from the corresponding author.

Do you need more time to submit your revision?

If you need an extension, [please contact us](#) and include your submission ID. You can find this ID in the revision request email or on the right-hand side of this page.

[Submit revision](#)

Do you need editing support?

If you have been asked to improve the language or presentation of your manuscript and would like the assistance of paid editing services, then Springer Nature Author Services can help. Our services include English language editing, manuscript formatting, figure preparation, translation, and more.

Find out more and get 15% off your order by visiting [our Author Services](#).

Please note that use of an editing service is neither a requirement nor a guarantee of publication. Free assistance is available from [our resources page](#).

Progress so far

Show history

1. Submission received
2. Technical check
3. Editorial assignment
4. With editor
5. Peer review

Learn [about our submission process](#)

Your submission

Title

Are Heat-Health Systems Socially Blind? Social Isolation and Dehydration-Related Healthcare Utilization Across Japan

Type

Research

Journal

International Journal of Biometeorology

Submission ID

0cd18650-d5db-481f-b00b-bbde3cda74f7

Submission version

v.1.1

Manuscript File[manuscript_main_IJB.docx](#)**Need help?**

If you have any questions about this submission, you can [email the Editorial Office](#) (francisanthony.vega@springernature.com).

For general enquiries, please look at our [support information](#).

How was your experience today?